

Midterm 2 - Solutions

QUESTION 1

Let $\{F_t\}$ be the Poisson process that counts the arrivals of frozen turkeys at the supermarket. Its rate is $\eta := 50$, on the day timescale. The weight of the i -th turkey is given by $W_i \sim \text{Exp}(\lambda)$, for $\lambda := 1/15$. We assume that $\{W_i\}_{i \in \mathbb{N}}$ are i.i.d. and that $\{F_t\}_{t \geq 0}, \{W_i\}_{i \in \mathbb{N}}$ are independent.

a) Each arrival is independently assigned one of two labels according to

$$X_i = \begin{cases} 1 & \text{if } W_i \geq 10 \\ 2 & \text{if } W_i < 10 \end{cases}$$

Note that by independence of $\{F_t\}, \{W_i\}$, the labels are independent of $\{F_t\}$ as well. In addition, the labelling occurs with the following probabilities, for all $i \in \mathbb{N}$.

$$p_1 := \mathbb{P}(X_i = 1) = \mathbb{P}(W_i \geq 10) = e^{-10\lambda} = e^{-2/3}$$

$$p_2 := \mathbb{P}(X_i = 2) = 1 - \mathbb{P}(X_i = 1) = 1 - e^{-2/3}$$

By the principle of thinning, the processes $\{F_t^{(1)}\}$ and $\{F_t^{(2)}\}$ that count the arrivals of each type are independent Poisson processes with rates $p_1\eta, p_2\eta$. We may thus compute the desired probability as

$$\mathbb{P}(F_t^{(1)} = 10) = \frac{e^{-p_1\eta} (p_1\eta)^{10}}{10!} = \frac{e^{-e^{-2/3}50} (e^{-2/3}50)^{10}}{10!}.$$

b) Let $\{C_t\}$ be the Poisson process that counts the arrivals of customers at the supermarket. Its rate is $\mu := 240$, on the day timescale (Note: the supermarket is open 24/7 and the hourly rate for customers is 10). Since by assumption $\{F_t\}_{t \geq 0}, \{C_t\}_{t \geq 0}$ are independent, the superposition principle allows us to conclude that $T_t := F_t + C_t$ is a Poisson process with rate $\eta + \mu$.

It is possible to generate the processes $\{F_t\}, \{C_t\}$ from $\{T_t\}$ via thinning by independently labelling each arrival of $\{T_t\}$ as a customer or a turkey with respective probabilities $\frac{\eta}{\eta+\mu}$ and $\frac{\mu}{\eta+\mu}$. By defining Y_i as the label of the i -th arrival of $\{T_t\}$, we may compute the desired probability as follows.

$$\begin{aligned} \mathbb{P}(\text{2nd customer does not get a turkey}) &= \mathbb{P}(Y_1 = \text{customer}, Y_2 = \text{customer}) \\ &\quad + \mathbb{P}(Y_1 = \text{turkey}, Y_2 = \text{customer}, Y_3 = \text{customer}) \\ &= \left(\frac{\mu}{\eta + \mu}\right)^2 + \left(\frac{\mu}{\eta + \mu}\right)^2 \left(\frac{\eta}{\eta + \mu}\right) \\ &= \left(\frac{240}{290}\right)^2 \left(1 + \frac{50}{290}\right) \end{aligned}$$

Grading comments:

- In both parts a and b: the independence assumptions required to apply thinning and superposition were not stated.

- In part a: instead of using thinning, some students computed $\mathbb{P}(F_1 = 10 \text{ and } W_i \geq 10, i = 1, \dots, 10)$, by relying on independence of $\{F_t\}_{t \geq 0}, \{W_i\}_{i \in \mathbb{N}}$. Note that this is not the correct interpretation of the question at hand.
- In part b: several students did not properly rescale the rate parameter on the day timescale. In a few cases, the parameter of the exponential weights was chosen as 15 rather than $1/15$, leading to a minor mistake in the final result. Several students incorrectly identified the event of interest.

QUESTION 2

- a) Let N denote the number of slices of the pie and D the fraction eaten by the dog. The modeling assumptions are $N \sim \text{Gamma}(2, 1/5)$ and $D \sim \text{Uniform}(0, 1)$ with N and D being independent. Denote by $M := N(1 - D)$ the number of slices that are left. The mean of M is

$$\begin{aligned}\mathbb{E}[N(1 - D)] &= \mathbb{E}[N] \mathbb{E}[1 - D] \quad (\text{by independence}) \\ &= \mathbb{E}[N] (1 - \mathbb{E}[D]) \quad (\text{by linearity of expectation}) \\ &= \frac{2}{1/5} \cdot (1 - \frac{1}{2}) = 5.\end{aligned}$$

Next, we calculate the variance using $\text{Var}[X] = \mathbb{E}[X^2] - \mathbb{E}[X]^2$.

$$\begin{aligned}\text{Var}[N(1 - D)] &= \mathbb{E}[N^2(1 - D)^2] - 5^2 \\ &= \mathbb{E}[N^2] \mathbb{E}[(1 - D)^2] - 25 \quad (\text{by independence})\end{aligned}$$

Notice that $1 - D$, the fraction of pie not eaten by the dog, is also uniform distributed on $[0, 1]$. We now calculate the second moments:

$$\begin{aligned}\mathbb{E}[N^2] &= \text{Var}[N] + \mathbb{E}[N]^2 = \frac{2}{(1/5)^2} + 10^2 = 150 \\ \mathbb{E}[(1 - D)^2] &= \mathbb{E}[D^2] = \text{Var}[D] + \mathbb{E}[D]^2 = \frac{1}{12} + \left(\frac{1}{2}\right)^2 = \frac{1}{3}\end{aligned}$$

$$\text{Hence, } \text{Var}[N(1 - D)] = 150 \cdot \frac{1}{3} - 5^2 = 25.$$

- b) The problem asks for $\mathbb{E}[N|M = 4]$. We can find it using the definition

$$\int_0^\infty x \cdot \mathbb{P}\{N \in dx | M = 4\}$$

or the formula

$$\int_0^\infty \mathbb{P}\{N > x | M = 4\} dx.$$

We first calculate

$$\mathbb{P}\{N \in dx | M = y\} = \frac{\mathbb{P}\{N \in dx, M \in dy\}}{\mathbb{P}\{M \in dy\}}$$

- To compute $\mathbb{P}\{M \in dy\}$, we first compute $\mathbb{P}\{M \leq y\}$: For $y > 0$,

$$\begin{aligned}
\mathbb{P}\{M \leq y\} &= \int_{\mathbb{R}} \mathbb{P}\{M \leq y|N = x\} \mathbb{P}\{N \in dx\} \\
&= \int_{\mathbb{R}} \mathbb{P}\left\{1 - D \leq \frac{y}{x} | N = x\right\} \mathbb{P}\{N \in dx\} \\
&= \int_{\mathbb{R}} \mathbb{P}\left\{1 - D \leq \frac{y}{x}\right\} \mathbb{P}\{N \in dx\} \quad (\text{by independence}) \\
&= \int_0^\infty \left(\frac{y}{x} \cdot 1_{[y, \infty)}(x) + 1_{(0, y)}(x)\right) \cdot \frac{\mu e^{-\mu x} (\mu x)^{2-1}}{(2-1)!} dx \\
&= \mu y \int_y^\infty \mu e^{-\mu x} dx + \mu^2 \int_0^y x e^{-\mu x} dx \\
&= \mu y e^{-\mu y} + \mu^2 \int_0^y x e^{-\mu x} dx.
\end{aligned}$$

Differentiating the cdf, we obtain

$$\mathbb{P}\{M \in dy\} = \mu e^{-\mu y} - \mu^2 y e^{-\mu y} + \mu^2 y e^{-\mu y} = \mu e^{-\mu y}.$$

We see that $M \sim \text{Exponential}(\mu)$.

- Next, we compute $\mathbb{P}\{N \in dx, M \in dy\}$. Applying Baye's formula, we write

$$\mathbb{P}\{N \in dx, M \in dy\} = \mathbb{P}\{M \in dy|N = x\} \mathbb{P}\{N \in dx\}.$$

For $0 < y < x$,

$$\mathbb{P}\{M \leq y|N = x\} = \frac{y}{x}.$$

We get $\mathbb{P}\{M \in dy|N = x\} = \frac{1}{x}$ for $y \in (0, x)$, i.e. $M|N = x \sim \text{Uniform}(0, x)$.

Hence,

$$\mathbb{E}[N|M = y] = \int_y^\infty x \cdot \frac{(1/x) \cdot \mu^2 x e^{-\mu x}}{\mu e^{-\mu y}} dx = \mu e^{\mu y} \int_y^\infty x e^{-\mu x} dx = y + \frac{1}{\mu}$$

Alternatively,

$$\begin{aligned}
\mathbb{E}[N|M = y] &= \int_0^y 1 dx + \int_y^\infty \int_x^\infty \frac{(1/z) \cdot \mu^2 z e^{-\mu z}}{\mu e^{-\mu y}} dz dx \\
&= y + e^{\mu y} \int_y^\infty e^{-\mu x} dx = y + \frac{1}{\mu}
\end{aligned}$$

Plugging in $\mu = 1/5$ and $y = 4$ gives $\mathbb{E}[N|M = 4] = 9$.

Grading comments:

- A common mistake was failing to express the random variable that is asked in the problem. Two examples are ND and $N - D$.
 - Fewer students calculate the variance correctly. For two independent random variables X and Y , we have $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, but this property is not true when calculating variances. Independence only allows us to separate the variance of a "sum" of random variables, i.e. $\text{Var}[X \pm Y] = \text{Var}[X] + \text{Var}[Y]$ if X and Y are independent.
- It is incorrect to argue that the answer is $4/\mathbb{E}[D] = 8$. No point was given if no other valid arguments were made.

- Partial credits were given to students who identified the correct conditional expectation or applied the correct formulas.

QUESTION 3

- a) We define the total number of bouts of indigestion at the end of hour t as S_t , and the number of bouts of indigestion during hour t as x_t . We know that the x_t 's are i.i.d., and

$$\mathbb{P}(x_t = 0) = \mathbb{P}(x_t = 1) = \mathbb{P}(x_t = 2) = \frac{1}{3}.$$

Also, we have

$$S_t = S_0 + x_1 + \cdots + x_t.$$

If we define T as "the time we decide to take some peptobismol", then

$$T := \min\{t \in \mathbb{N} | S_t \geq 3\}.$$

Then for question (a), our goal is to find $\mathbb{P}(T = \infty | S_0 = 0)$. We define

$$r(i) := \mathbb{P}(T = \infty | S_0 = i).$$

Then by first step analysis, we have the following equations:

$$r(0) = \frac{1}{3}r(0) + \frac{1}{3}r(1) + \frac{1}{3}r(2)$$

$$r(1) = \frac{1}{3}r(1) + \frac{1}{3}r(2) + \frac{1}{3}r(3)$$

$$r(2) = \frac{1}{3}r(2) + \frac{1}{3}r(3) + \frac{1}{3}r(4)$$

Notice that by definition, $r(3) = r(4) = 0$, since the random walk will immediately stop when $S_0 \geq 3$. Then solve these equations, we get $r(0) = 0$, i.e., $\mathbb{P}(T = \infty | S_0 = 0) = 0$.

- b) We use the same notations in (a). Then in (b), we want to find $\mathbb{E}[T | S_0 = 0]$. We define

$$h(i) := \mathbb{E}[T | S_0 = i].$$

Then by first step analysis, we have the following equations:

$$h(0) = 1 + \frac{1}{3}r(0) + \frac{1}{3}r(1) + \frac{1}{3}r(2)$$

$$h(1) = 1 + \frac{1}{3}r(1) + \frac{1}{3}r(2) + \frac{1}{3}r(3)$$

$$h(2) = 1 + \frac{1}{3}r(2) + \frac{1}{3}r(3) + \frac{1}{3}r(4)$$

Notice that by definition, $h(3) = h(4) = 0$, since the random walk will immediately stop when $S_0 \geq 3$, which means $T = 0$ given $S_0 = 3$ or $S_0 = 4$. Solving these equations, we get $h(0) = \frac{27}{8}$, i.e., $\mathbb{E}[T | S_0 = 0] = \frac{27}{8}$.

Grading comments:

- For (a), some students use Murphy's Law, but it cannot be applied here directly.
- For both subquestions, most students manage to set up the problem correctly, but many were not able to reach the correct answer because they either made a mistake in the first step analysis, the derivation of the system or mistakes in solving the system.